

PUBLIC TRANSPORTATION AND ACCESS TO GROCERY STORES IN OKLAHOMA CITY

Caleb Karr

```
## Module 1 - Create new feature layer with only transit stops with ntd_ids of 60017 and 60018.

# Import arcpy and set the workspace
import arcpy
import os
arcpy.env.workspace = ("C:/Users/ckarr1/Downloads/Project/NTAD_National_Transit_Map_Stops_863312227489020878")

# Set in and out features and where clause
# in_features is the shapefile we are using to narrow down to only select stops occurring in Oklahoma City.
# The only field in Map_Stops that we can use to narrow the selection in ntd_id. The two ntd_ids used in OKC are 60017 and 60018,
# However, Tulsa also uses these ids, so we will eventually remove Tulsa's transit stops.
# out_features is where we are going to put the new shapefile along with its name
# where_clause is where we select that we want the new shapefile to only include transit stops with ntd_ids of 60017 or 60018
National_Transit_Map_Stops = "National_Transit_Map_Stops.shp"
ok_bus_stops = ("C:/Users/ckarr1/Downloads/Project/NTAD_National_Transit_Map_Stops_863312227489020878/ok_bus_stops.shp")
where_clause = "ntd_id = '60017' or ntd_id = '60018'"

# Lastly, we run the Select function, adding in and out features and where clause, which creates the new feature layer
arcpy.analysis.Select(National_Transit_Map_Stops, ok_bus_stops, where_clause)
```

```
## Module 2

# We need to run code that selects only Oklahoma County
# Set in feature (COUNTY_BOUNDARY) and where you would like the out feature class to be located, and its name
# Where clause tells the code what condition needs to be met to be selected.

COUNTY_BOUNDARY = "COUNTY_BOUNDARY"
ok_county = arcpy.env.workspace = "C:/Users/ckarr1/Downloads/Project/NTAD_National_Transit_Map_Stops_863312227489020878/ok_county"
where_clause = "COUNTY_NAM = 'OKLAHOMA'"

arcpy.analysis.Select(COUNTY_BOUNDARY, ok_county, where_clause)
```

```
## Module 3
# We need to join ok_county.shp to the NTAD National Transit Map Stops folder.
# Set the workspace to this folder because it is where ok_county (which we are moving) is located
arcpy.env.workspace = "C:/Users/ckarr1/Downloads/Project/county"

# We are using this code to join ok_county and put it into the NTAD National Transit Map Stops folder so we can clip them next
out_feature = os.path.join(r"C:/Users/ckarr1/Downloads/Project/NTAD_National_Transit_Map_Stops_863312227489020878/ok_county.shp")
arcpy.CopyFeatures_management(ok_county, out_feature)
```

```
## Module 4
# We need to run code that clips only the bus stops occurring in Oklahoma County
# The in feature is the bus stops shapefile and we are clipping those stops with the Oklahoma County boundary, leaving only the bus stops occurring the county
ok_county_transit_stopsClip = "ok_county_transit_stopsClip.shp"
arcpy.analysis.Clip(ok_bus_stops, ok_county, ok_county_transit_stopsClip)
```

```
## Module 5
# We are now creating a 0.5 mile buffer from each bus stop in Oklahoma County

# Import in and out features, and buffer distance, then run code
ok_county_transit_stopsBuffer = "ok_county_transit_stopsBuffer.shp"
buffer_distance_or_field = "0.5 miles"
arcpy.analysis.Buffer(ok_county_transit_stopsClip, ok_county_transit_stopsBuffer, buffer_distance_or_field)
```

```
## Module 6
# Clip Oklahoma census tracts to the Oklahoma County boundary
#Set in, clip, and out features, then run code
ok_tracts = "cb_2018_40_tract_500k"
ok_county_census_tracts = "ok_county_census_tracts.shp"

arcpy.analysis.Clip(ok_tracts, ok_county, ok_county_census_tracts)
```

```
## Module 7
# We need to select the top 10 highest populated census tracts in Oklahoma County
# Here we are listing the tracts individually
top_10_tracts = [
    "1085.11",
    "1074.03",
    "1081.06",
    "1085.23",
    "1069.03",
    "1074.01",
    "1072.12",
    "1070.01",
    "1081.09",
    "1083.18"]

# Here we are setting our in and out features and our where clause. The format of the where_clause code is important
# to make sure we are clear with what tracts we want to select
in_features = ok_county_census_tracts
out_feature_class = arcpy.env.workspace = ("C:/Users/ckarr1/Downloads/Project/NTAD_National_Transit_Map_Stops_863312227489020878/top_10_tracts.shp")
where_clause = ("NAME IN ('1085.11', '1074.03', '1081.06', "
"'1085.23', '1069.03', '1074.01', "
"'1072.12', '1070.01', '1081.09', '1083.18')")

arcpy.analysis.Select(in_features, out_feature_class, where_clause)
```

```
## Module 8
# This is a user-input section.
# Users will be able to put their census tract in and see if they are living in one of the
# Top 10 most populated tracts or not

# Define the function and list the tracts that we would like to define as a top 10 tract
top_10_tracts = {
    "1085.11",
    "1074.03",
    "1081.06",
    "1085.23",
    "1069.03",
    "1074.01",
    "1072.12",
    "1070.01",
    "1081.09",
    "1083.18"
}
#This function tells the user to enter their tract number
user_tract = input("Enter your Census Tract Number: ")

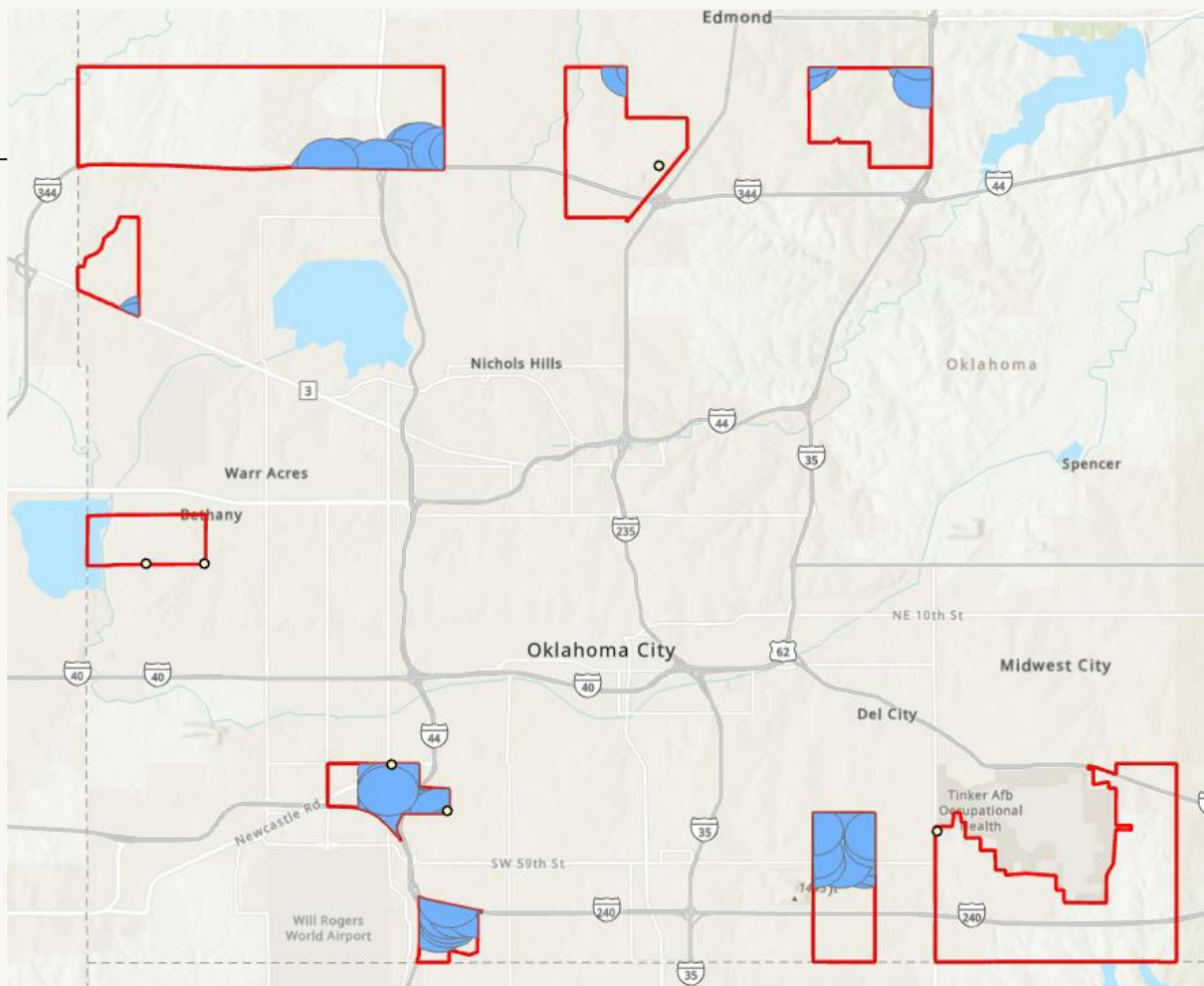
# This conditional statement determines what is printed based on user input
if user_tract in top_10_tracts:
    print("This tract is one of the Top 10 most populated census tracts in Oklahoma County.")
else:
    print("This tract is not one of the Top 10 most populated census tracts in Oklahoma County.")
```

```
## Module 9
# Clip buffered bus stops to the Top 10 population tracts in Oklahoma County
# Set in, clip, and out features, then run code

top_10_tracts_bus_stopsBuffer = "top_10_tracts_bus_stopsBuffer.shp"
arcpy.analysis.Clip(ok_county_transit_stopsBuffer, top_10_tracts, top_10_tracts_bus_stopsBuffer)
```

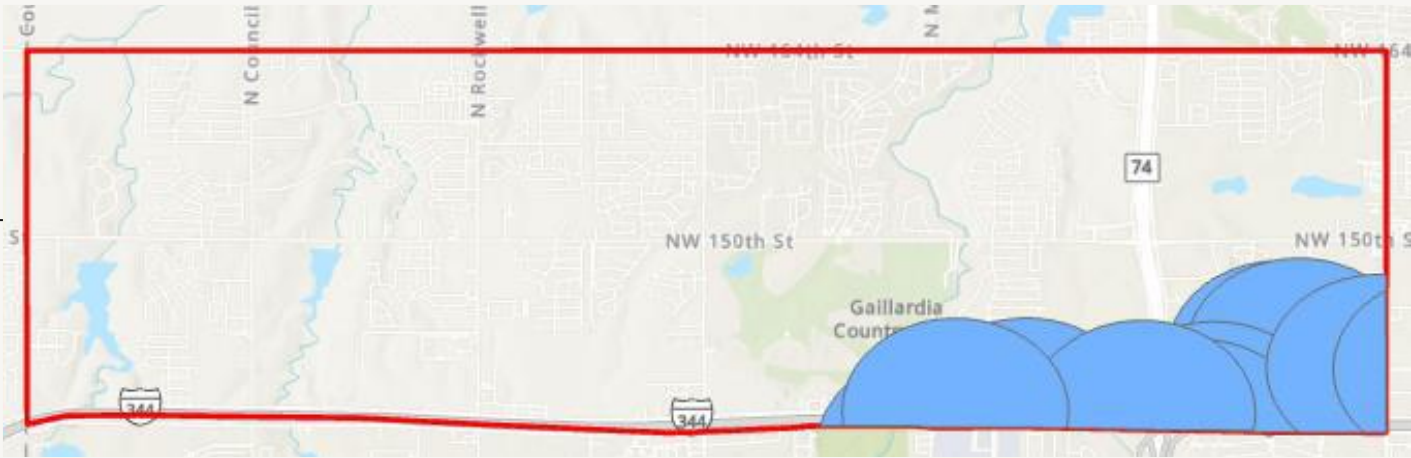
```
## Module 10
# Import a table with XY coordinates of grocery stores within Top 10 population tracts

# Add the location of the csv file with the coordinates,
# then add the name of the new feature you are creating.
# Next, make sure your csv file has the coordinates properly formatted and that
# the titles of the columns are labelled "Longitude" and "Latitude".
# Lastly, the coordinate system needs to be set so the code knows where to put the points.
arcpy.management.XYTableToPoint("C:/Users/ckarr1/Downloads/grocery_store_coords.csv",
                                "grocery_stores",
                                "Longitude", "Latitude",
                                coordinate_system = arcpy.SpatialReference(4326))
```



Top 10 Census Tracts for population in OKC with buffered (.5 miles) bus stops

Census tract 1085.11



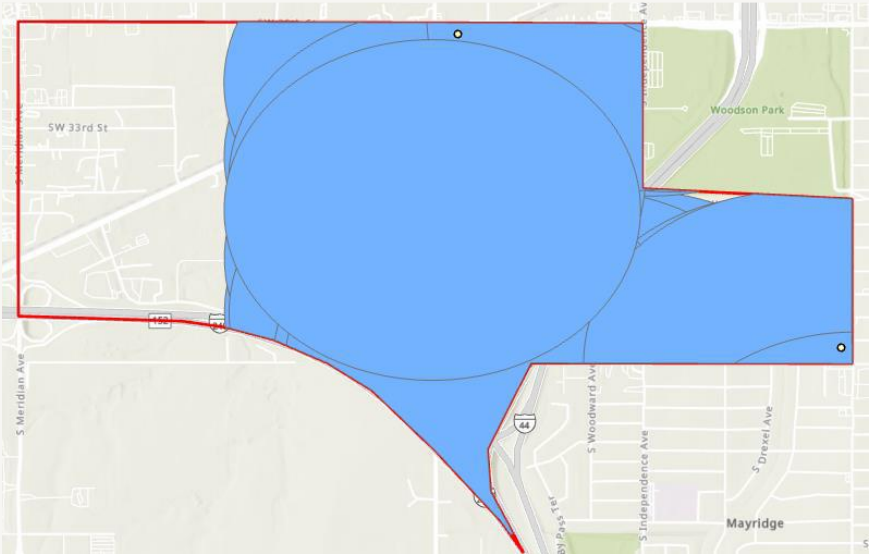
Census tract 1085.23



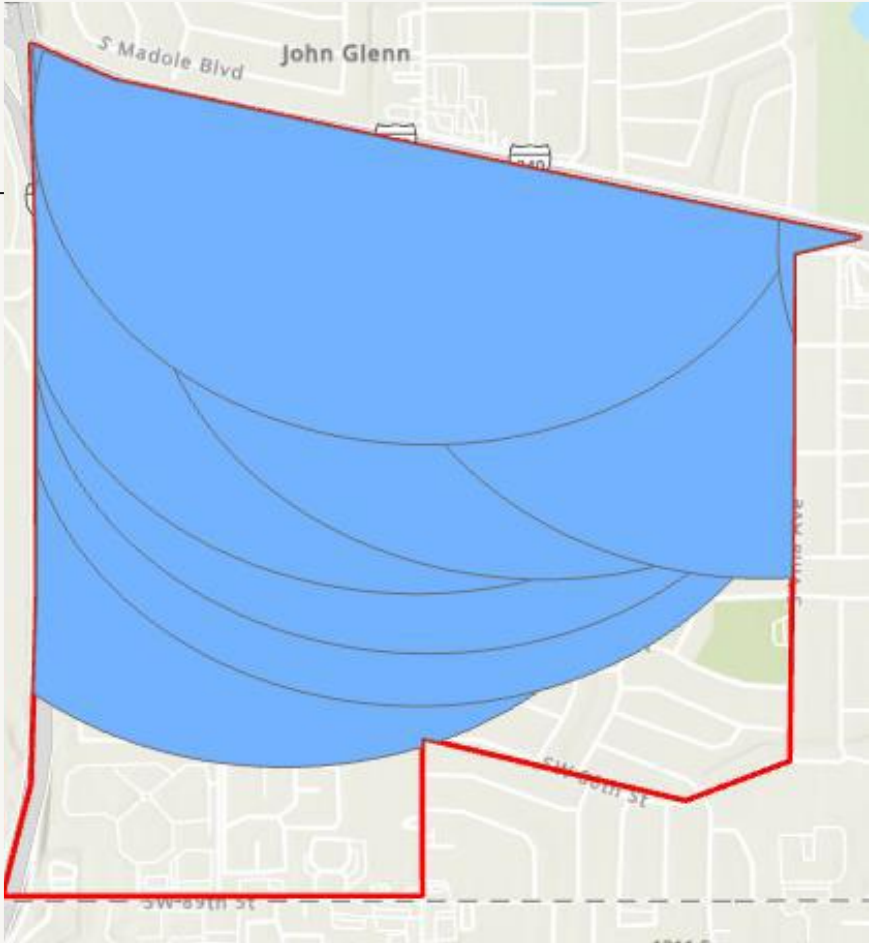
Census tract 1069.03



Census tract 1070.01



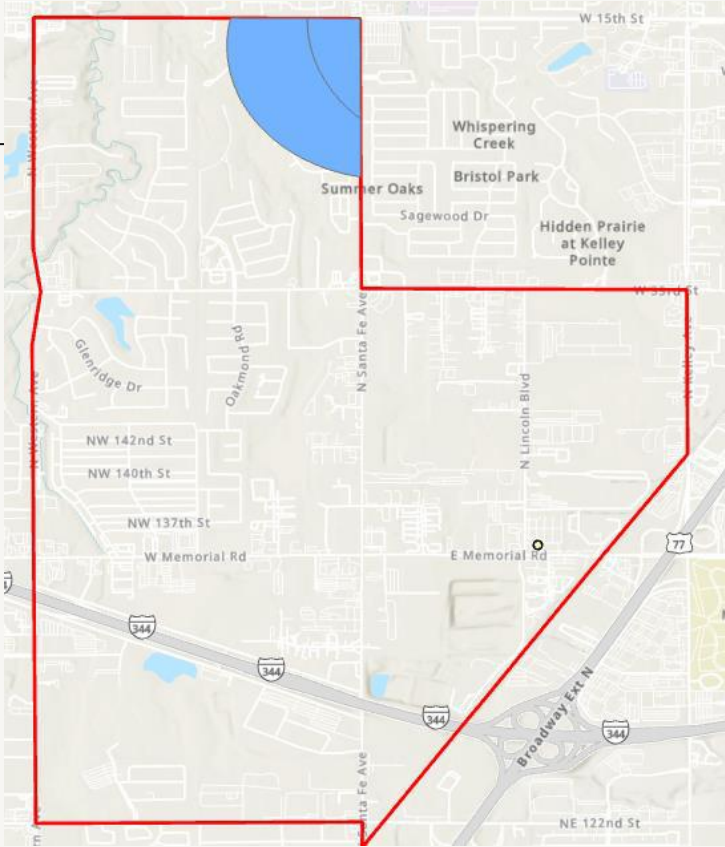
Census tract 1072.12



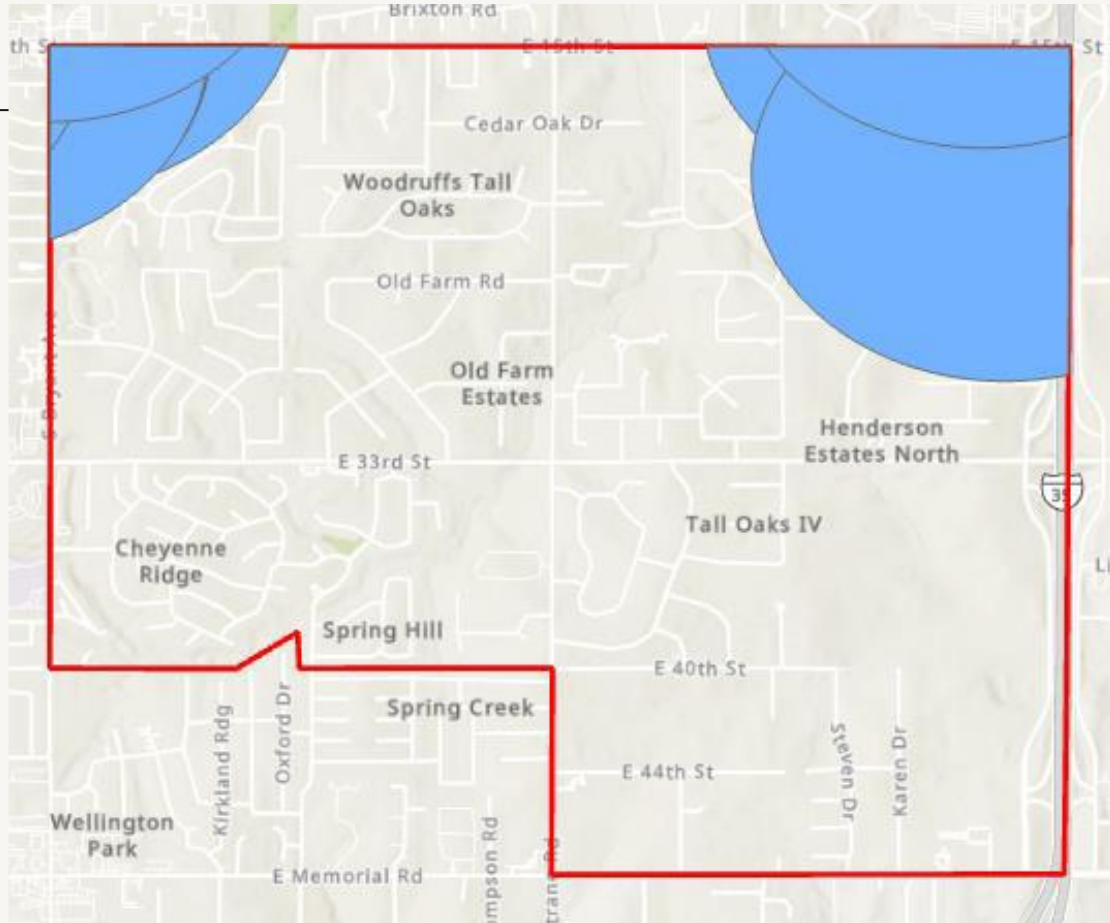
Census tract 1074.01



Census tract 1074.03



Census tract 1081.09



Census tract 1081.06

